

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

MATHEMATICS
PAPER 1

4004/1
2 hours 30 minutes

JUNE 2026 SESSION

Additional materials:
Mathematical Instruments
CETV

INSTRUCTIONS TO CANDIDATES

Write your Name, Centre number and Candidate number in the spaces at the top of each page.
Check that all the pages are in the booklet and ask the invigilator for a replacement if there are duplicate or missing pages.
If working is needed for any question, it must be shown in the space below that question.
Omission of essential working will result in loss of marks.
Decimal answers which are not exact should be given correct to three significant figures unless stated otherwise.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.
This paper is marked out of 100.

This question paper consists of 21 printed pages.

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Answer **all** questions.

**NEITHER MATHEMATICAL TABLES NOR SLIDE RULES
NOR CALCULATORS MAY BE USED IN THIS PAPER.**

- 1 Express 31,6824 correct to
- (a) 3 significant figures,

Answer (a)

[1]

- (b) 2 decimal places,

Answer (b)

[1]

- (c) the nearest ten.

Answer (c)

[1]

- 2 (a) Convert $10\ 011_2$ to a number in base 5.

Answer (a)

[2]



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(b) Simplify $42_5 + 13_5$, giving the answer in base 5.

Answer (b)

[1]

3 Set S is such that

$S = \{\text{square, rectangle, equilateral triangle, kite, trapezium, parallelogram}\}$

From the set S, write down the name of the shape which has

(a) rotational symmetry of order 3,

Answer (a)

[1]

(b) rotational symmetry of order 2 and exactly 2 lines of symmetry,

Answer (b)

[1]

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(c) one line of symmetry.

Answer (c)

[1]

4 Express $\frac{1}{4x - 2y} - \frac{y}{4x^2 - y^2}$ as a single fraction in its simplest form.

Answer

[3]

5 A model greenhouse is made to a scale of $\frac{1}{200}$.

(a) Calculate the length of the model in, cm, if the actual length of the greenhouse is 7,8 m.

Answer (a)



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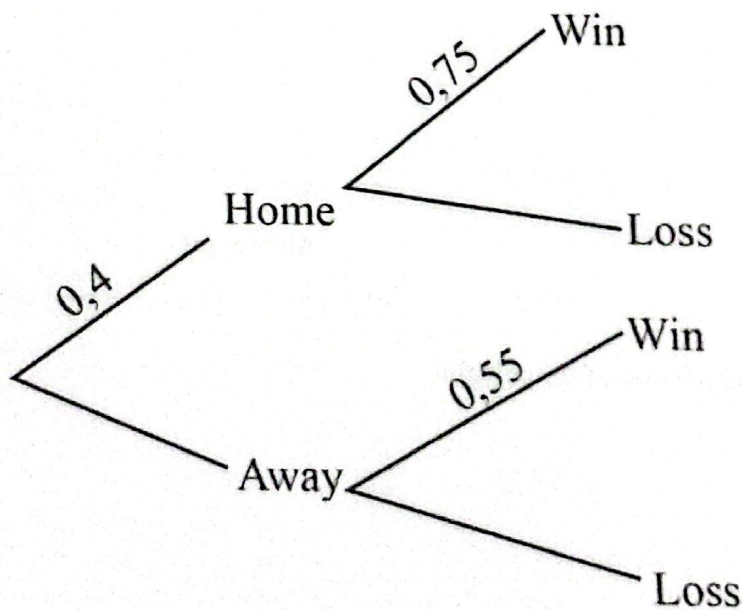
- (h) On the model, the base area is 70cm^2 . Calculate the actual base area of the greenhouse, giving the answer in m^2 .

Answer (b)

[2]

- 6 A rugby team plays 40% of its matches at home. They win 75% of their home matches and only 55% of their away matches.

- (a) Complete the tree diagram below.



Answer (a) on the tree diagram

[2]

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- (b) Find the probability that the team wins a match.

Answer (b)

[2]

- 7 An n -sided regular polygon is such that its interior angle is 5 times the exterior angle.

Calculate the

- (a) size of each exterior angle,

Answer (a)

[2]

- (b) number of sides of the polygon.

Answer (b)

[2]



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- 8 (a) Solve the inequality $-5 < 2x + 3 < 1$. Give the solution in the form $a < x < b$; where a and b are integers.

Answer (a)

[3]

- (b) Write down the largest integer, x , which satisfies $-5 < 2x + 3 < 1$.

Answer (b)

[1]

- 9 Given that $m = 6 \times 10^3$ and $n = 8 \times 10^6$, evaluate

(a) $\frac{m}{n}$,

Answer (a)

[2]

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(b) $\sqrt[3]{n}$.

Answer (b)

[2]

- 10 (a) By selling a drink for \$78,00 a shopkeeper makes a profit of 30%.
Find the price at which the shopkeeper should sell it to make a profit of 25%.

Answer (a)

[3]

- (b) Value Added Tax (VAT) at the rate of 15%, is added to the marked prices of all articles in a shop.
An article is marked \$60. Find the price of the article including VAT.

Answer (b)

[2]

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11 (a) Simplify

(i) $2\frac{1}{4} - 1\frac{2}{3}$

Answer (a)(i)

[2]

(ii) $\frac{1}{3} \div \frac{4}{5}$

Answer (a)(ii)

[2]

(b) Express $\frac{25}{40}$ as a percentage.

Answer (b)

[1]

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- 12 A full water tank had 575 litres of water left when a farmer used 3,865 kilolitres of water on irrigation.

(a) Find the capacity of the tank when full.

Answer (a)

[3]

- (b) A borehole pump, pumps 7200 litres of water in 1 hour. Find the time taken, in minutes, to fill up the tank when empty.

Answer (b)

[2]



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13 Given that $y = \frac{100}{x^2}$,

(a) State the type of variation represented by $y = \frac{100}{x^2}$,

Answer (a)

[1]

(b) Calculate

(i) y when $x = 5$,

Answer (b)(i)

[2]

(ii) x when $y = 4$.

Answer (b)(ii)

[2]

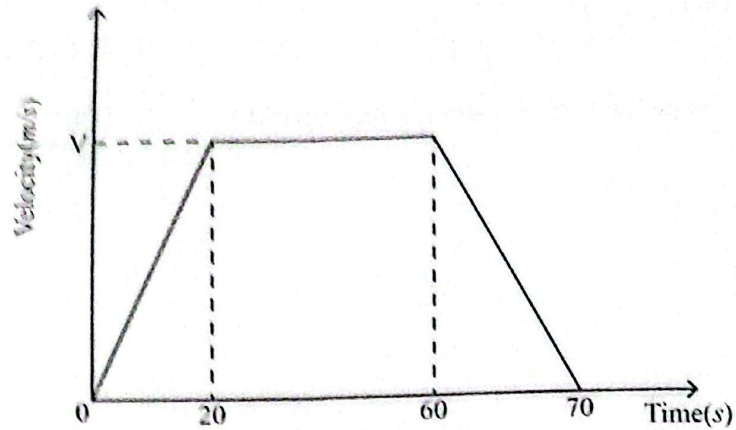
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The diagram is the velocity - time graph of a train which travels a distance of 880 metres in 70 seconds.

Calculate the

- (a) value of V ,

Answer (a) _____

[2]

- (b) acceleration during the first 20 seconds.

Answer (b) _____

[2]



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- (c) retardation during the last 10 seconds.

Answer (c)

[1]

- 15 If $f(x) = 6 - x$,
find

- (a) $f(2)$,

Answer (a)

[2]

- (b) the value of x for which $f(x) = x$.

Answer (b)

[3]

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- 16 Two cylindrical buckets are similar. The larger bucket has a height of 30cm and base diameter 16 cm. The smaller bucket has base diameter of 8cm.

(a) Calculate the height of the smaller bucket.

Answer (a)

[2]

- (b) The volume of the smaller bucket is 754 cm^3 .
Calculate the volume of the larger bucket, in cm^3

Answer (b)

[3]

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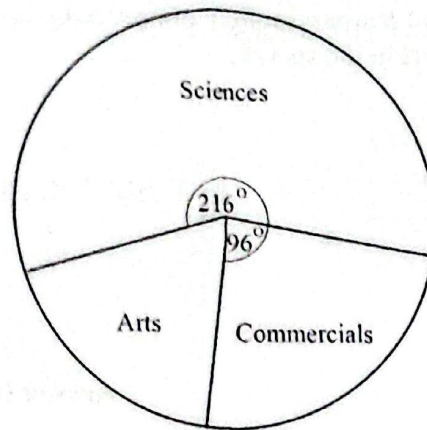
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17



In a survey, some A level learners were asked which of the three categories of subjects, Commercials, Sciences and Arts, they were studying. The results are represented in a pie chart above.

- (a) Calculate the percentage of learners who study Arts,

Answer (a)

[2]

- (b) Find in its simplest form, in the form $m:n$, where m and n are integers, the ratio of the number of learners who study Sciences to those who study Commercials.

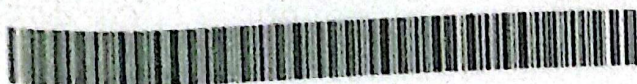
Answer (b)

[1]

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- (c) Given that 44 learners study Commercials, calculate the number of learners who took part in the survey.

Answer (c)

[2]

- 18 A triangle has sides of length 6cm and 5cm. The angle between these two sides is 120° .

Use as much of the information given below as is necessary.

[$\cos 60^\circ = 0,5$; $\sin 60^\circ = 0,866$; $\tan 60^\circ = 1,73$]

Calculate the

- (a) area of the triangle,

Answer (a)

[2]

- (b) length of the third side, leaving the answer in surd form.

Answer (b)

[3]



- 19 Given that $A(2; 2)$, $B(0; k)$ and $C(3; 2)$ are points on a plane, relative to the origin, O, find,

(a) $|\vec{OA}|$, giving the answer in surd form,

Answer (a)

[2]

(b) $\vec{BC}$ in terms of k ,

Answer (b)

[1]

(c) k given that $\vec{OA}$ is parallel to $\vec{BC}$.

Answer (c)

[3]



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20 (a) Solve the equation $8^{x+1} = 32$.

Answer (a)

[3]

(b) Simplify $\frac{\log 36}{\log_6 \frac{1}{6}}$

Answer (b)

[3]



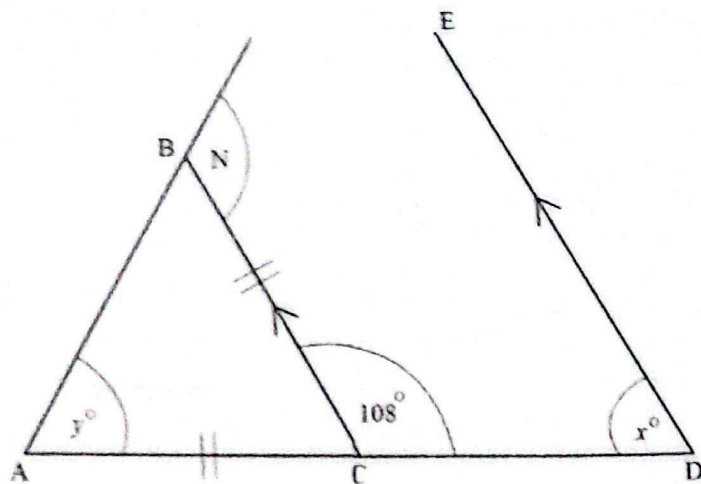
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The diagram shows triangle ABC in which $AC = BC$.

Point D is on AC produced and DE is parallel to CB and $\angle BCD = 108^\circ$.

Calculate the angle marked

(a) x ,

Answer (a)

[2]

(b) y ,

Answer (b)

[2]



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(c) N.

Answer (c)

[2]

- 22 A bag contains x red balls and 8 green balls. The probability of picking a red ball is $\frac{3}{5}$.

(a) Find the probability of picking a green ball.

Answer (a)

[1]

(b) Calculate the number of red balls in the bag.

Answer (b)

[3]



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- (c) If two balls are picked at random one after the another without replacement, find the probability that they are of the same colour,

Answer (c)

[2]

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